

Dr. Mariana Jaber

Cosmologist



About me

I am a cosmologist working on the interface between theory and observations, linking theoretical models with the computational analysis of astronomical data.

Personal information

Name: Mariana Jaber Bravo
Nationality: Mexican

Areas of specialization

Theoretical and Observational Cosmology · Large Scale Structure · Peculiar Velocities · Cosmic web · Galaxy surveys · Dark Energy and Modified gravity · N-body simulations

Referee

Modern Physics Letters A
Physical Review Letters
Physical Review D
Astronomy & Astrophysics
The Astrophysical Journal
MSCA Expert (2025, 2026)

Membership in Collaborations

WST (Active)
CLUES (Active)
LSST-DESC (Seeking Rights)
DESI (2018-2019)
eBOSS (2014-2019)

Community Service

Supernova Foundation: Senior Council (since 2026), Leadership Board (2023-2026), Mentor (since 2020)
CFT-PAS: Gender Equality Plan (2023)
SMF & INALI: Member of the Editing Team (2017)
UV-Peraj: Tutor (2009-2010), Directive Board (2010-2012)

Languages

Spanish (native)
English (proficient)
French (intermediate)
Polish (basic)

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[MarianaJBr](https://www.marianajaber.com)
www.marianajaber.com

RESEARCH EXPERIENCE

April 2026– Present day	National Centre for Nuclear Research PRINCIPAL INVESTIGATOR · Warsaw, PL
Sept 2021– March 2026	Center of Theoretical Physics, Polish Academy of Sciences POSTDOCTORAL RESEARCHER (3 YEARS) PRINCIPAL INVESTIGATOR (1 YEAR) · Warsaw, PL
Sept 2019– Aug 2021	Nicolaus Copernicus University (Postdoctoral Researcher) FACULTY OF PHYSICS, ASTRONOMY AND INFORMATICS · Toruń, PL

TEACHING EXPERIENCE

8-22 Oct 2021	Short specialised courses COMPUTATIONAL COSMOLOGY GROUP WORKSHOP · Świeradów-Zdrój, PL "Multipoles of BOSS DR12 LRG catalog" & Themed Discussion Sessions
Apr, Nov, 2017	Short specialised courses CINVESTAV · CDMX, MX "Constraining cosmological parameters with Cosmic Clocks" & "Use of N-body kit"
2014-2016	Lecturer PHYSICS I · Curso UNAMCDMX, MX 6-8 groups of 30+ students per semester, 4 semesters
2015-2016	Teaching Assistant PHYSICAL COSMOLOGY · Faculty of Sciences, CDMX, MX 14 students per class, 2 semesters

EDUCATION

Jan, 2019	PhD, summa cum laude PHYSICS · IF-UNAM
July, 2014	M.Sc. PHYSICS · IF-UNAM
March, 2011	B.S., summa cum laude PHYSICS · Fac. of Physics, UV

THESIS

2025-2029 Director of PhD thesis (NCBJ, Warsaw, PL)
2025-2027 Director of Master thesis (ongoing) (Masaryk Univ., Brno, CZ)
2024-2025 Director of bachelor thesis (UW, Warsaw, PL)
2023-2024 Director of master thesis (UJ, Kraków, PL)
April 2024 - Jury member. Master in Astrophysics (UNAM, MX)
July 2019 - Jury member. Master in Astrophysics (UNAM, MX)

AWARDS & GRANTS

2024	NCN - SONATA grant
2023	CFT - Young Scientists Grant.
2019 - 2023	SNI (Candidate)
2018	ARC-SDSS Travel grant

ACADEMIC EVENTS

Research visits	6
Seminars	14
Conferences	18 (contributions) & 23 (participation)
Outreach talks	6
Popular Sci.	"The inheritance of economic inequality" (published)
Book chapter	Programming exercises in Python "PRAXIS. Física I". MacMillan Editorial, ISBN:9786075403052
Papers	10 Published (8 as first or second author)

EVENTS ORGANISED

2026	VISTULA (40 participants)
2024	Symposium on alternative theories of gravity (20+ participants)
2022	II Roman Juskiewicz Symposium (65+ participants)
2017	Hands-on Python workshop CinvesPy (70+ participants)
2017	Num & Stat Methods in Cosmology (40+ participants)